

Project Proposal: The Algorithmic Mirror

An AI-Driven Framework Exploring Mental Health, Digital Habits, and Academic Performance in Gen Z

Prepared For:	Academic & Institutional Stakeholders	Date:	July 11, 2026
Focus Area:	Predictive Analytics & Digital Wellness	Status:	Proposal / Specifications v1.0

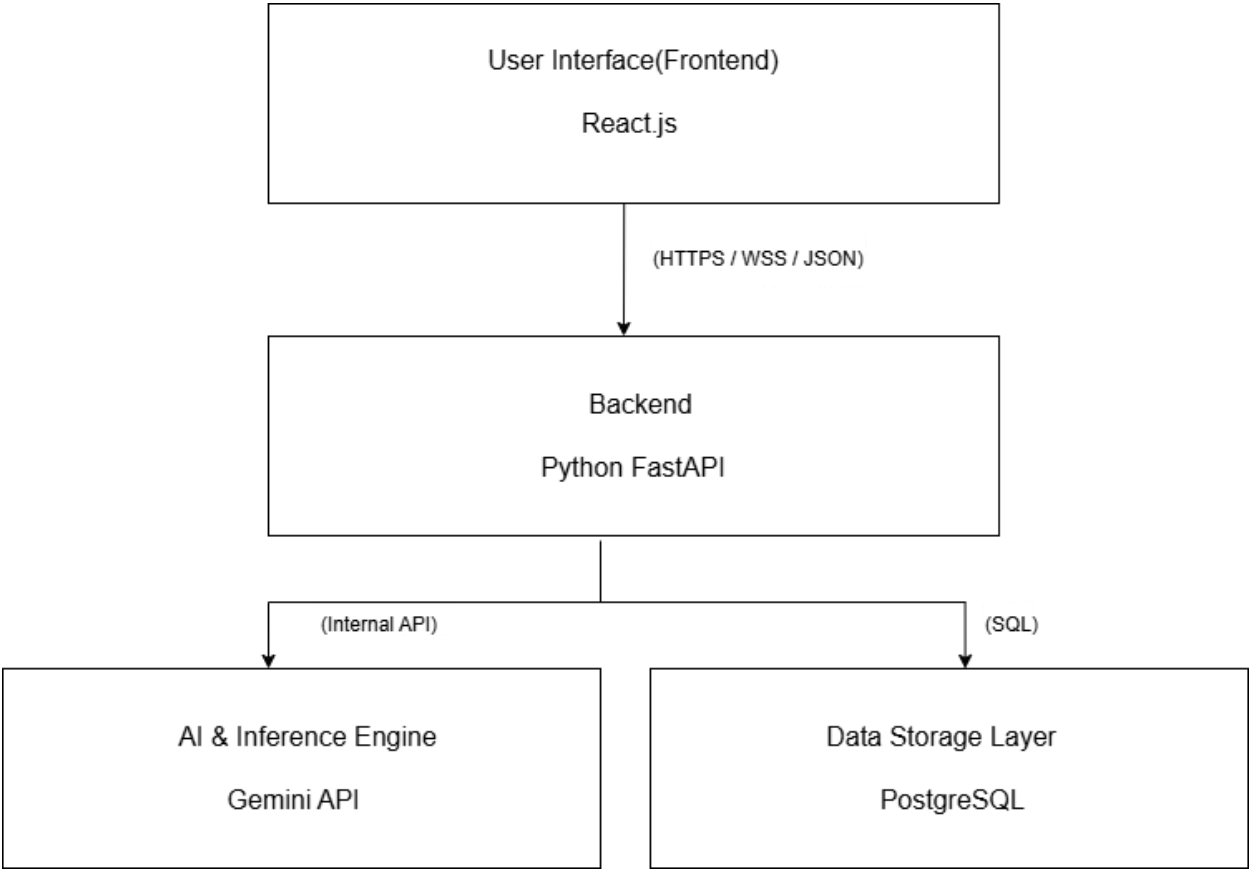
Overview

- The Algorithmic Mirror is a data-driven initiative designed to investigate and mitigate the complex interplay between digital lifestyles and psychological well-being within Generation Z. As the first generation to grow up entirely in the era of smartphones and ubiquitous social media, Gen Z faces unprecedented cognitive and emotional challenges. This project develops an intelligent system utilizing a robust behavioral dataset of 1,000 unique users containing metrics on stress, anxiety, depression, platform screen-time distribution and sleep efficiency.

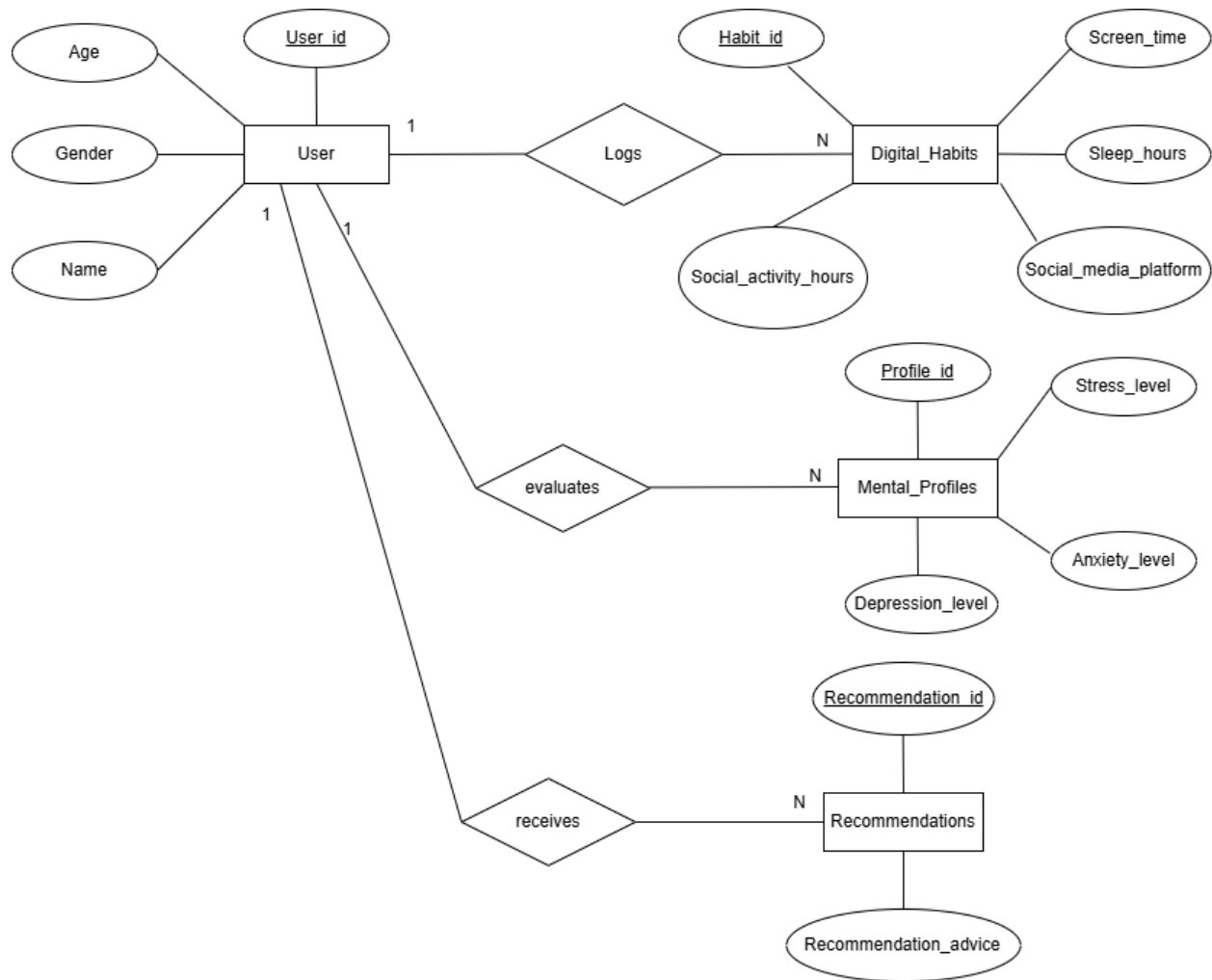
Challenges Intended to Solve Using AI Integration

- Finding Complex Relationships:** Traditional analysis struggles to identify how different digital habits, such as late-night social media use or excessive screen time, affect mental health. AI can analyze large amounts of data to discover hidden patterns and relationships.
- Early Risk Prediction:** Instead of identifying mental health issues after they occur, AI can predict potential risks by analyzing users' digital behavior. This allows users to take preventive actions before their mental well-being or academic performance declines.
- Personalized Recommendations:** General advice like "use your phone less" is often ineffective. AI provides personalized recommendations based on each user's digital habits and mental health profile, making the advice more relevant and practical.

System Architecture Diagram



System ER Diagram



Functional and Non-Functional Requirements

Functional Requirements

1. User Registration and Login
2. User Profile Management
3. Behavioral Data Collection
4. Mental Health Prediction
5. Risk Categorization

Non-Functional Requirements

- 1. Performance
- 2. Security
- 3. Reliability
- 4. Usability
- 5. Data Integrity
- 6. Privacy

Technology Details

Layer	Technology Component
Frontend	React.js
Backend Framework	Python, FastAPI
Machine Learning Core	Gemini API
Database Engine	PostgreSQL

Conclusion

The Algorithmic Mirror translates unstructured behavioral and psychological patterns into predictive, actionable wellness pathways. Then the system categorize user’s mental health level as high, medium, low and give useful advices as the output.

Github Link

<https://github.com/shalithawijesekara117/The-Algorithmic-Mirror.git>